

ChurnIQ

Bank Customer Attrition Risk Intelligence

A Random Forest churn prediction pipeline, deployed as a live decision-support tool

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github.com/yemifatodu/bank-churn-predictor

What we'll cover

01

The business problem

Why churn prediction matters in retail banking

02

The data

10,000 customers, 13 attributes, zero missing values

03

What the data reveals

Five findings, including two genuinely counterintuitive ones

04

The model

Random Forest pipeline, performance, and an honest limitation

05

From model to product

A live, explainable decision-support app

06

What's next

A concrete, prioritized roadmap

Banks find out about churn too late

By the time an account closes, the retention window has already passed. Customer data — balance, tenure, product holdings, activity — carries predictive signal about who's at risk, but that signal rarely reaches a relationship manager before the customer leaves.

Why it matters

- Acquiring a new customer costs materially more than retaining one
- Retention teams operate reactively without a forward-looking score
- Blanket retention campaigns waste budget on low-risk customers
- A risk score lets banks target spend at the highest-risk, highest-value accounts

OBJECTIVES

What this project set out to do

- 1 Predict** individual customer churn probability from demographic and account attributes available at scoring time
- 2 Explain** which attributes drive that risk — not just produce a black-box score
- 3 Deploy** the model as a tool a non-technical user can actually operate, not a notebook
- 4 Evaluate honestly** at the class level, given known imbalance in the churn outcome

THE DATA

10,000 customers, no cleanup required

10,000

Customer records

13

Attributes + target

0

Missing values

20.37%

Overall churn rate

Attributes used by the model

Credit Score

Geography

Gender

Age

Tenure

Balance

Products Held

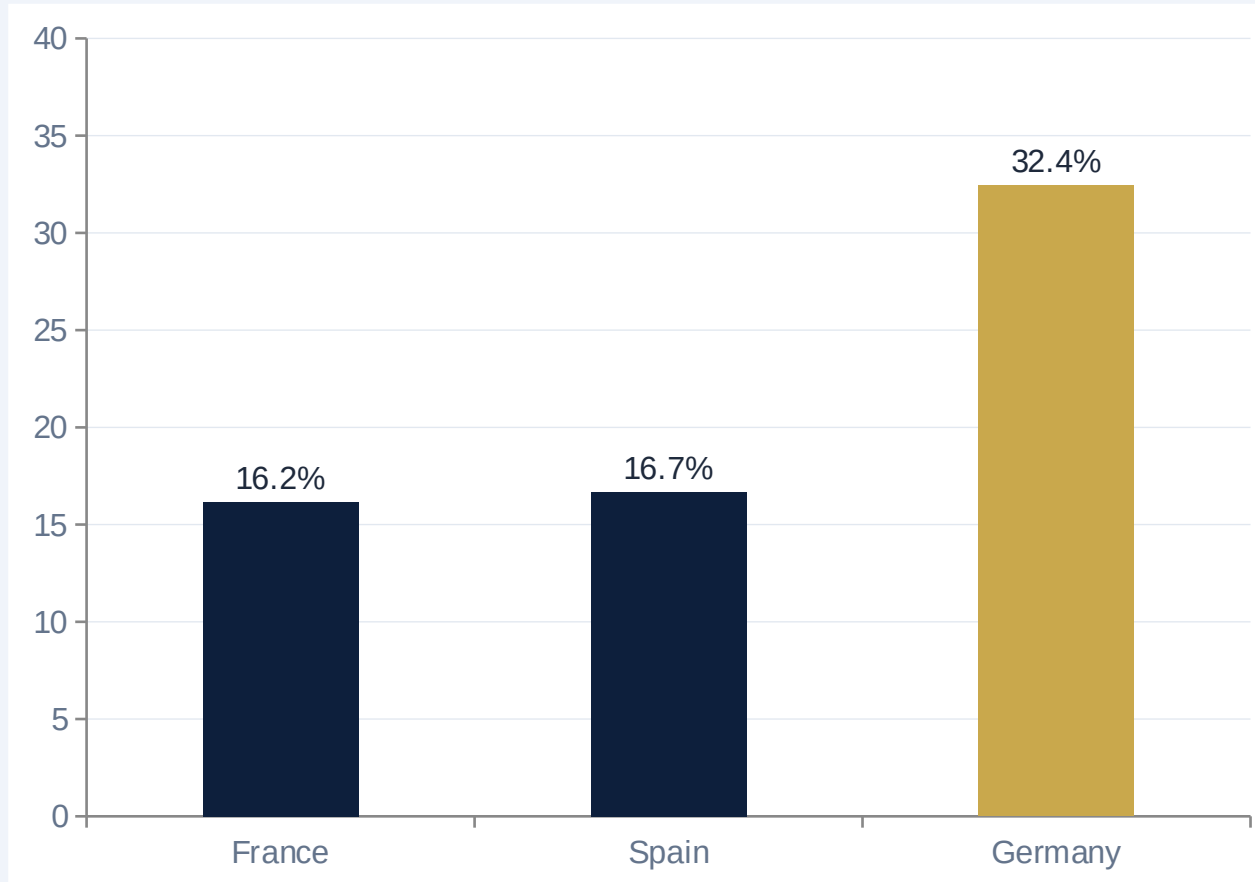
Has Credit Card

Active Member

Estimated Salary

FINDING 01

Germany churns at twice the rate of France or Spain



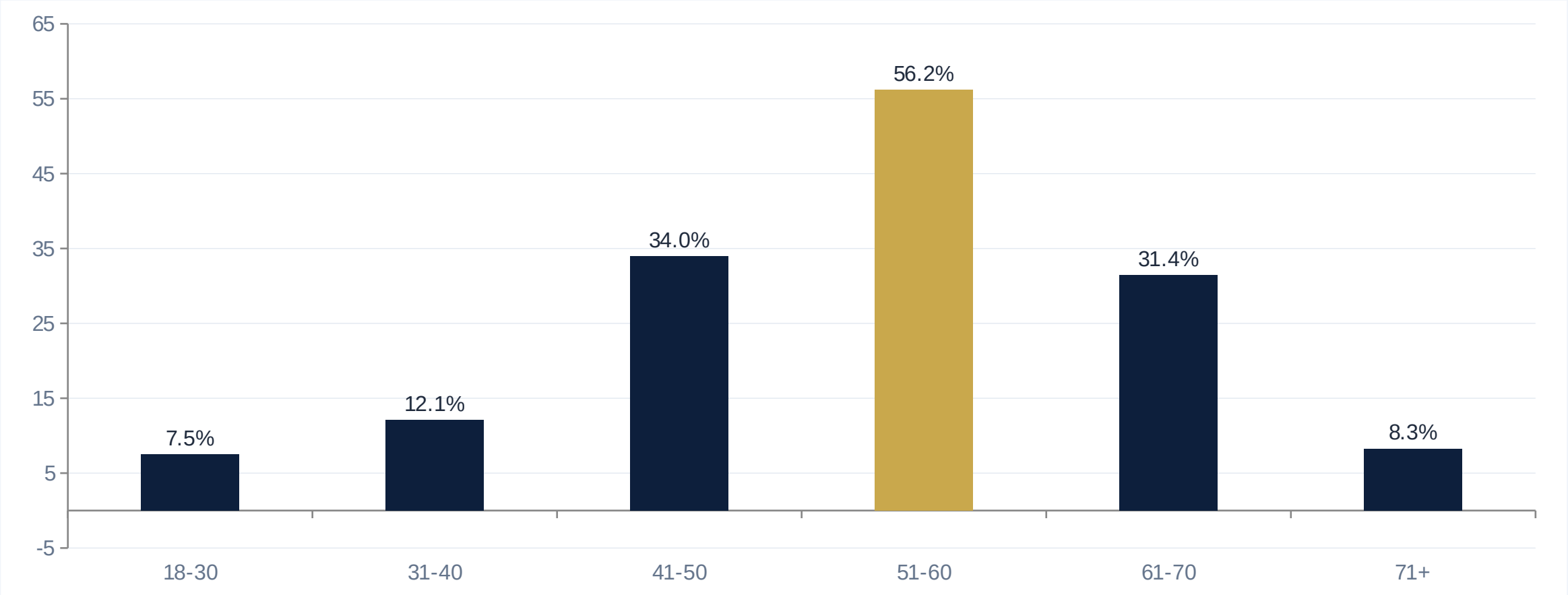
2,509

German customers in the dataset — a similar base size to Spain (2,477), but double the churn rate.

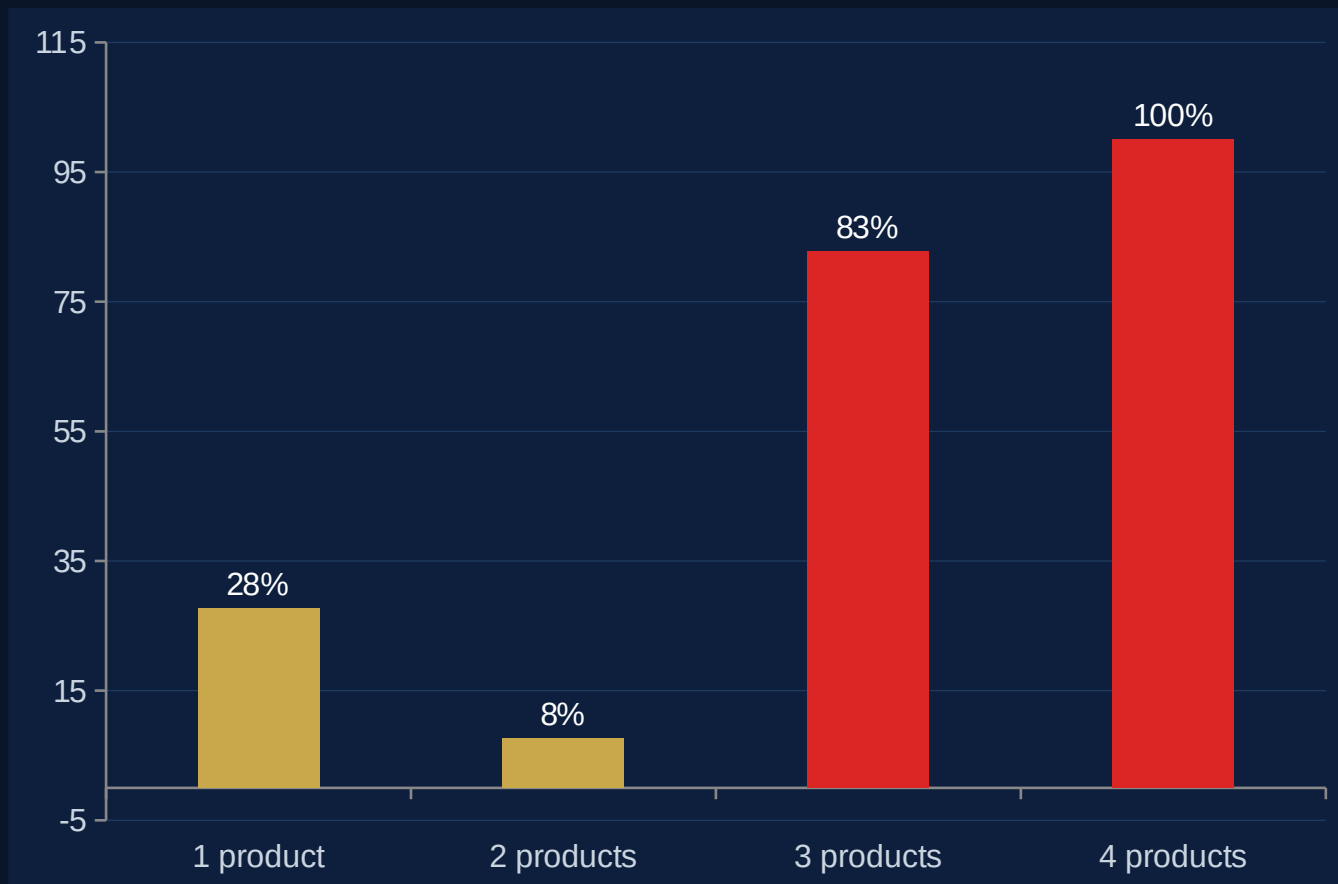
Root cause isn't identifiable from this data alone — competitive pressure, product fit, and service quality are all plausible and merit direct investigation before assuming a single explanation.

FINDING 02

Churn risk peaks sharply in the 51–60 age band



More products held means far higher churn, not lower



Why this matters

This is a U-shape, not a straight line — which is why the simple linear correlation between product count and churn is nearly zero (-0.048), even though it's the model's #2 predictor.

326 customers hold 3–4 products (out of 10,000) but churn at 83–100%. Likely over-bundling or forced sign-ups shortly before exit — a priority segment to investigate before scaling cross-sell.

A non-zero balance is a risk signal, not a loyalty signal

BY ACCOUNT BALANCE

13.82%

Zero balance
(3,617 customers)

24.08%

Non-zero balance
(6,383 customers)

Nearly double the churn rate for customers holding a balance — the opposite of what 'stickier high-value customer' intuition would predict.

BY ACTIVITY STATUS

14.27%

Active member

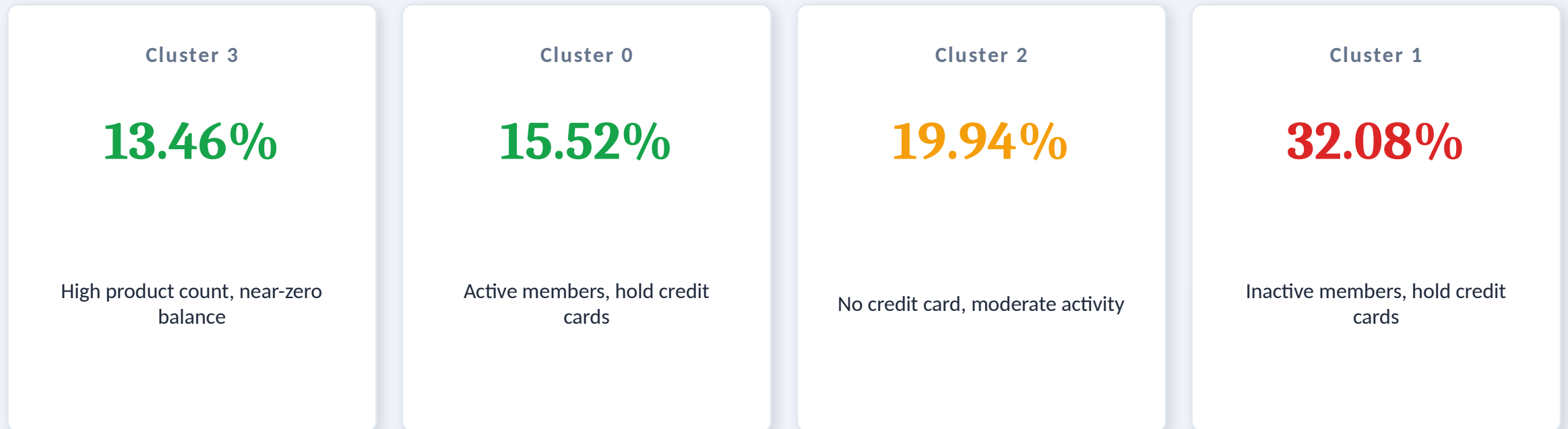
26.85%

Inactive member

A clear, directionally usable gap — and a much lower-friction retention lever (engagement campaigns) than trying to change balance behavior.

Customer segmentation reveals a 2.4x risk gap

KMeans clustering (k=4, chosen via elbow + silhouette analysis) on 8 behavioral features, visualized with PCA:



Cluster 1 — inactive credit-card holders — churns at more than double Cluster 3's rate. A coarser signal than the individual model, but immediately actionable as a targeting list. Silhouette scores were modest (0.12–0.15): real clusters, not sharply separated.

From raw data to a deployed decision tool



Single stratified 80/20 split (random_state=42). No k-fold cross-validation and no hyperparameter search yet — a fixed, manually-chosen configuration.

A single, well-scoped Random Forest classifier

HYPERPARAMETERS

n_estimators	200
max_depth	10
min_samples_split	5
min_samples_leaf	2
random_state	42

PREPROCESSING & FEATURES

- 8 numeric features standardized with StandardScaler
- Geography and Gender one-hot encoded (drop='first')
- CustomerId and Surname excluded as non-predictive IDs
- No engineered features yet — raw attributes only

These are the deployed hyperparameters. The original notebook used an unconstrained Random Forest (default depth, non-stratified split) — this configuration was a deliberate later refinement.

RESULTS

86.80% accuracy, 0.864 ROC-AUC

86.80%

Accuracy

0.864

ROC-AUC

0.83

Churn Precision

0.44

Churn Recall

CONFUSION MATRIX • n = 2,000

	Pred. Stay	Pred. Churn
Actual Stay	1,556	37
Actual Churn	227	180

READ THIS CAREFULLY

227 of 407 churners in the test set were missed. High accuracy alone hides this — it's only visible by looking at the churn class specifically. This is the model's real limitation, addressed next.

Note: an earlier, unconstrained model iteration scored 0.70 precision / 0.47 recall / 0.8456 AUC. These numbers reflect the refined, deployed version.

Accuracy looks strong. Recall tells the real story.

79.6% of customers in this dataset don't churn. A model that mostly predicted “stays” would already land near 80% accuracy — so 86.80% is a real improvement, but it isn't, by itself, proof the model is catching at-risk customers.

0.83

PRECISION
When flagged, trust it

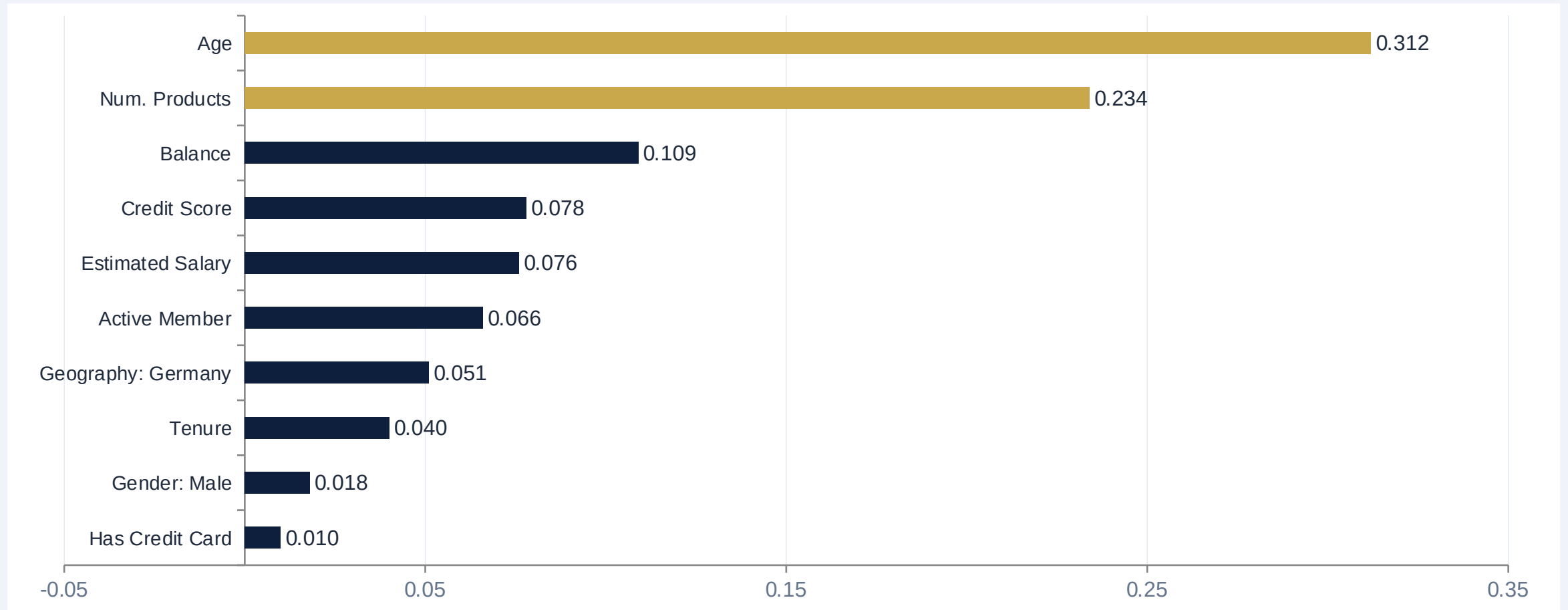
0.44

RECALL
Misses over half of churners

WHAT THIS MEANS IN PRACTICE

- The model is conservative — it under-flags rather than over-flags
- Fine for a first pilot; not yet reliable as the sole basis for retention budget decisions
- Clear, known fixes exist: class weighting, threshold tuning, resampling

What actually drives the model's predictions



A live, explainable decision-support tool

Risk verdict

Churn / Retain call with a probability score and confidence, color-coded for quick scanning

Feature drivers

Per-prediction breakdown of which inputs pushed the verdict, rendered live in-app

Action plan

Distinct, pre-written retention actions for high-risk customers vs. cross-sell/loyalty actions for low-risk ones

RECOMMENDATIONS

Where to act, in order of confidence

High confidence

Pilot an inactive-member engagement campaign — large, directionally clear effect, low implementation risk

High confidence

Stop treating high balance as a loyalty signal in manual retention triage — the data runs the other way

Needs investigation

Audit the 3–4 product segment and the Germany market qualitatively before scaling any related campaign

Before scaling

Close the churn-recall gap (0.44 today) before using this model as the sole basis for budget decisions

A concrete, prioritized roadmap

1

Improve churn-class recall — class weighting, threshold tuning, or resampling

2

Add a comparison model (e.g. gradient boosting) rather than relying on one algorithm

3

Introduce k-fold cross-validation and a real hyperparameter search

4

Layer in SHAP-based, per-customer explainability

5

Engineer features the EDA already points to — product-count flag, age bands, balance-to-salary ratio

6

Add automated tests, CI, and a documented license

ChurnIQ

Built by Yemi Fatodu — Data Scientist · BI Specialist · Full-Stack Product Builder

Open to freelance & contract engagements

Live demo: yemifatodu-bank-churn-predictor.streamlit.app

Code: github.com/yemifatodu/bank-churn-predictor

Portfolio: yemifatodu.online · LinkedIn: linkedin.com/in/yemifatodu